Estimating Undocumented Populations from ACS Data

United States, Texas, and Texas health-insurance status, 2008–2024

Draft analytic memo • July 2, 2026

Bottom Line

Draft notes

This memo implements the residual method of Warren, Warren, and Zheng (2023) in a reduced form. The residual method starts from a simple accounting definition, $U_t = C_t - L_t$: the undocumented population equals survey-measured noncitizens C_t (undercount-adjusted) minus the legal noncitizen population L_t, reconstructed from administrative records (lawful admissions, naturalizations, life-table mortality, and emigration). Because L_t moves slowly, the ratio $\theta_t = U_t/C_t = 1 - L_t/C_t$ — the undocumented share of noncitizens — is a slow-moving demographic quantity. We take C_t from ACS table B05001 and *pin* θ_t to the values implied by published residual-method releases at anchor years — Pew Research Center, the Migration Policy Institute (MPI), and the Center for Migration Studies of New York (CMS — the immigration research institute, not the federal Medicare agency) — linearly interpolating θ_t between anchors. This reproduces each source's published U_t exactly at its anchor years and lets observed ACS noncitizen movement carry between-anchor variation. Following the residual-method literature, "undocumented" includes people with temporary or liminal protections (DACA, Temporary Protected Status, humanitarian parole, pending asylum); Pew reports that over 40 percent of the 2023 unauthorized population held some protection from deportation. The construction is useful for levels and cross-source sensitivity but is not a direct ACS tabulation of legal status (see Section 2.1).

- The Pew/MPI residual-method vintage puts the 2024 U.S. undocumented population at about **14.9 million**, or **4.4 percent** of Americans. Warren/CMS's directly published 2024 residual estimate is 14.6 million.
- The same Pew/MPI vintage puts the 2024 Texas undocumented population at about **2.10 million**, or **6.7 percent** of the Texas population.
- For Texas health insurance, the estimate is about **1.34 million uninsured** and **0.76 million insured** undocumented Texans in 2024, or **64 percent uninsured** and **36 percent insured**. The uninsured undocumented population equals about **4.3 percent of all Texans**.
- A parallel implementation of the same residual method using the CMS/Warren residual-method series (Warren/CMS anchors) gives a higher Texas 2024 count of about **2.32 million** (**7.4 percent** of Texans), **1.39 million uninsured (60 percent)**, and **0.93 million insured**. The two independent residual-method vintages land within about 9 percent of each other for Texas in 2024 (about 4 percent in 2023), which is the most robust convergence finding in this memo.
- In 2023 the main Pew/MPI-anchored Texas estimate is **inside the CMS/Pew/MPI residual-method range**: our Texas anchor is 1.97 million, compared with CMS at 2.05 million, Pew at 2.05 million, and MPI at 1.97 million.

- **The main contribution is a reusable method, not just these numbers.** The residual shortcut runs from published tables for any subgroup, geography, or policy domain with an anchor and a survey count; the health-coverage results here are one narrower application. The largest single uncertainty is which source anchors the estimate: across states the three published sources disagree by a median of about 14 percent, so a state number often needs a range, not a point.

1 Data and Definitions

Acronyms. ACS is the American Community Survey. CMS is the Center for Migration Studies of New York, an immigration research institute — throughout this memo, CMS never refers to the Centers for Medicare & Medicaid Services. MPI is the Migration Policy Institute. DHS is the U.S. Department of Homeland Security.

The residual method. All three sources whose values anchor this workflow (Pew, MPI, CMS) produce their undocumented counts using variants of the same residual accounting $U_t = C_t - L_t$, where C_t is survey-measured noncitizens and L_t is the legal noncitizen population reconstructed from DHS lawful-admission records, life-table mortality, naturalizations, and emigration (Warren, Warren, and Zheng, 2023; Passel and Cohn 2018; Warren 2014).[1] Our workflow does not rebuild the DHS/life-table/emigration components from scratch; it interpolates $\theta_t = U_t/C_t$ between years where one of these residual-method sources has published U_t, keeping $U_t = C_t - L_t$ exact at anchor years by construction. This is the reduced-form residual method described in Section 2.1.

Who counts as undocumented. All three residual-method sources define the unauthorized (undocumented) population to include unauthorized entrants and visa overstayers *plus* people with temporary or liminal protections: DACA recipients, Temporary Protected Status holders, humanitarian parolees, and pending asylum applicants. This matters for interpretation — Pew estimates that more than 40 percent of the 2023 population had some protection from deportation, and CMS reports that liminal-status immigrants were more than a third of its 2024 total.

Do the three sources define the same population? Largely yes, which is worth showing so the cross-source spread is not mistaken for a definitional artifact. All three count the temporary and liminal statuses as unauthorized and exclude lawful permanent residents (Table 1). What differs is the *magnitude* each assigns to some groups — Pew counts about 0.70 million parolees in the 2023 population, CMS about 1.09 million — so part of the spread across sources is compositional, not a pure difference in method.

[1] Robert Warren, John Robert Warren, and Ping Zheng, "A New Residual Approach for Estimating Undocumented Populations," *International Migration Review*, DOI 10.1177/01979183231195280 (2023). The Warren et al. (2023) contribution generalizes earlier residual methods (Warren and Passel 1987; Warren 2014) by *deriving* the year-specific emigration rate directly from current-year ACS movement rather than fixing it from 1980–82 estimates. That is precisely the innovation this workflow leans on when it lets ACS noncitizen movement carry between-anchor variation.

Table 1: How the three residual-method sources classify each group in their undocumented total

Group	Pew	CMS	MPI	Why it matters
Entered without inspection	In	In	In	Core undocumented population
Visa overstays	In	In	In	Core undocumented population
DACA recipients	In	In	In	Temporary protection, not permanent status
TPS holders	In	In	In	Designations shift 2023–2024 totals
Humanitarian parolees	In	In	In	Grew sharply after recent parole programs
Pending asylum applicants	In	In	In	Largest liminal group (~2.6–3.6M)
Lawful permanent residents	Out	Out	Out	Legal noncitizen stock (L), subtracted

Source: "In" = counted in the undocumented total; "Out" = excluded. Pew treats *granted* asylees/refugees as lawful in the year of admission; only *pending* applicants are counted here. Sources: Pew 2025 report and methodology; CMS State and National Data Tool ("About the data"); MPI methodology for assigning legal status; all accessed July 2026 (URLs in Reproducibility).

ACS inputs. The workflow uses ACS 1-year detailed tables pulled by Stata's `getcensus` package:

- **B05001, Nativity and Citizenship Status**: total population, naturalized citizens, and noncitizens.
- **B27020, Health Insurance Coverage Status and Type by Citizenship Status**: noncitizen insured, private coverage, public coverage, and uninsured counts.

The ACS health-insurance table is available in this workflow beginning in 2009. Population estimates start in 2008. Standard ACS 1-year estimates were not released for 2020, so the panel skips 2020.

Two parallel residual-method vintages, same method, different anchors. We produce two implementations of the residual method side by side, using different published anchor sets:

- **Pew/MPI vintage (main).** θ_t is pinned at Pew's published U.S. residual-method estimates (2010–2023) and at MPI's Texas profile values (2019, 2023). Pew's national number is a residual-method estimate produced from ACS/CPS data.[2] MPI's Texas number is also a residual construction from pooled ACS data with SIPP-based status imputation.[3]
- **Warren/CMS vintage.** θ_t is pinned at CMS's published residual-method estimates: Warren's Journal on Migration and Human Security series for the United States

[2] Pew Research Center, *U.S. Unauthorized Immigrant Population Reached a Record 14 Million in 2023*, August 21, 2025. Pew's methodology page states that the residual approach compares a demographic estimate of lawful immigrants with total immigrants measured in ACS or CPS data.

[3] Migration Policy Institute, *Profile of the Unauthorized Population: Texas*, accessed July 2, 2026. MPI's Texas 2023 profile uses pooled 2019–2023 ACS data and SIPP-based status imputation, weighted to a 2023 unauthorized immigrant population estimate.

(2010–2023) plus the 2024 State and National Data Tool release, and CMS's Texas state-by-year values for 2016, 2018, 2023, and 2024. The Warren et al. (2023) construction — ACS noncitizens minus a year-by-year rolled-forward legal population with ACS-derived emigration — *is* the workflow's methodological reference point.[4]

Neither series is a "benchmark check" of the other or an "ACS fill-in" of a fixed 2023 number. Both are residual-method estimates that reproduce their source's published U_t exactly at anchor years and use the same observable ACS C_t between anchors. The 2023 cross-source spread (CMS 12.24M, Pew 14.00M, MPI 13.74M nationally; CMS 2.05M, Pew 2.05M, MPI 1.97M in Texas) is a range of independent residual-method releases.[5] For 2024, CMS's direct national count is 14,611,000; direct Texas count 2,317,000, including 1,385,000 uninsured and 932,000 insured undocumented Texans.

What is not directly rebuilt. This workflow does not independently reconstruct DHS lawful-admission records, life-table survival rates, ACS-derived emigration, or ACS undercount adjustments. Those elements are the substance of Warren et al.'s (2023) new residual method. Here they enter *indirectly*, through the published Pew, MPI, and CMS/Warren U_t values that we import as anchors. The trade-off is intentional: the reduced-form residual method inherits the anchor sources' full residual-accounting for free at anchor years, at the price of not rebuilding the demographic components in between. Section 2.1 discusses when this approximation is safe and when a full year-by-year Warren reconstruction would be preferred.

2 Methods

2.1 Reduced-Form Residual Method: Interpolating θ_t Between Published Anchors

Warren accounting. The residual method as formalized by Warren, Warren, and Zheng (2023) begins from

$$U_{g,t} = C_{g,t} - L_{g,t},$$

where $C_{g,t}$ is survey-measured noncitizens for geography g in year t (in this workflow, ACS table B05001, undercount-adjusted by the anchor sources), and $L_{g,t}$ is the legal noncitizen population rolled forward year by year from administrative components:

$$L_{g,t} = L_{g,t-1} + \text{arrivals}_{g,t} - \text{naturalizations}_{g,t} - \text{deaths}_{g,t} - \text{emigration}_{g,t}.$$

Warren et al.'s contribution is to *derive* the year-specific emigration term from current-year ACS movement rather than fixing it from historical (1980–82) estimates.

[4] CMS State and National Data Tool, `undoc_2024.csv`, accessed July 2, 2026; CMS, *Understanding the U.S. Undocumented Population: New 2024 Estimates from CMS*, May 5, 2026.

[5] Immigration Research Initiative, *50 States: Immigrants by Number and Share*, November 10, 2025, side-by-side reporting of CMS, Pew, and MPI 2023 state estimates.

Why the ratio is slow-moving. Define

$$\theta_{g,t} \equiv \frac{U_{g,t}}{C_{g,t}} = 1 - \frac{L_{g,t}}{C_{g,t}}.$$

Because $L_{g,t}$ is a stock that moves slowly (arrivals, naturalizations, deaths, and emigration are all small fractions of L in any single year), and because $C_{g,t}$ also moves slowly, $\theta_{g,t}$ is a slow-moving demographic quantity. In this workflow, the U.S. Pew θ ranges from 0.52 in 2010 to 0.60 in 2023 — a smooth path over thirteen years. This is why linear interpolation of θ between years with credible published U values is a reasonable approximation.

Reduced-form implementation. Every year in our panel, we observe $C_{g,t}$ directly from ACS. At each *anchor year* a where a residual-method source has published $P_{g,a}$, we set

$$\theta_{g,a} = \frac{P_{g,a}}{C_{g,a}},$$

and between anchors we linearly interpolate θ; before the first and after the last anchor we hold it flat at the nearest anchor's value. The estimate is then

$$\widehat{U}_{g,t} = \theta_{g,t} \times C_{g,t}.$$

By construction $\widehat{U}$ reproduces every anchor source's published value exactly at its anchor years, and ACS noncitizen movement ($C_{g,t}$, which enters both the definitional equation and the multiplicative step) carries between-anchor variation. Two vintages of the same construction run in parallel, using different residual-method sources for their anchors:

Vintage	Published anchor values
U.S. Pew	2010: 11.4M; 2015: 11.0M; 2017: 10.5M; 2019: 10.2M; 2021: 10.5M; 2022: 11.8M; 2023: 14.0M
U.S. Warren/CMS	2010: 11.7M; 2019: 10.4M; 2021: 10.3M; 2022: 10.9M; 2023: 12.2M; 2024: 14.6M
Texas MPI	2019: 1,739,000; 2023: 1,966,000
Texas Warren/CMS	2016: 1,597,000; 2018: 1,730,000; 2023: 2,052,500; 2024: 2,317,000

Both vintages use the same ACS $C_{g,t}$; they differ only in whose residual-method U is imported at anchor years. Neither is a "check" or a "fill-in" of the other — they are two independent applications of the same reduced-form Warren accounting.

The residual lawful population. Rearranging, $\widehat{L}_{g,t} = C_{g,t} - \widehat{U}_{g,t}$. At anchor years this exactly equals the residual lawful count the source is (implicitly) using; between anchors it inherits the interpolated θ.

Where this workflow deviates from a full Warren reconstruction. The full year-by-year construction of Warren et al. (2023) would take DHS lawful admissions by class and cohort, apply life-table survival to L, derive emigration from the current-year ACS drop, and adjust arrivals and C for age-, sex-, and Hispanic-origin-specific ACS undercount. This workflow does none of that directly: the arrivals, naturalizations, mortality, emigration, and undercount adjustments are inherited from the anchor sources' own residual constructions, which θ then averages between. That trade-off buys the ability to run the full time series with two Stata commands and eight numbers, at the cost of not exposing (or being able to sensitivity-test) the components. Direct implementation of the full Warren recursion is a natural next step (Section 4); the accompanying Stata package (`stata_package/residualundoc.ado`) supports both the reduced-form interpolation used here and the direct year-by-year variant.

Caveats. Three cautions follow from the reduced form. First, because the anchor sources scale C up for ACS undercount before dividing by their reconstructed L, our θ embeds those undercount adjustments; the estimate is not literally "the undocumented share of the raw ACS noncitizen count." Shares of the (unadjusted) ACS total population are therefore approximate. Second, two tails are extrapolations rather than published values: the U.S. 2008–2009 years extend the 2010 anchor backward, and the Texas 2008–2018 years extend the 2019 MPI anchor backward (the Texas 2010 extrapolation lands near 1.64M, close to Pew's Texas estimate of the period, which is reassuring but not a published MPI number). Third, part of the sharp 2022-to-2023 rise reflects benchmark *methodology* updates — Pew reweighted its 2022–2023 series and CMS introduced a new methodology in its 2024 release — so change measured across that break mixes real growth with definitional change.

2.2 Validation at Anchor Years

Because $\theta_{g,t}$ is pinned to each source's published residual-method value at anchor years, Figure 1 shows the estimated lines passing exactly through the published Pew (U.S.) and MPI (Texas) markers, and `anchor_validation.csv` records a zero deviation at every anchor. The interpolation is visible between anchors: the U.S. Pew vintage falls from 11.4M (2010) to 10.2M (2019), matching Pew's published decline, then rises to 14.0M by 2023 and an extrapolated 14.9M in 2024. This replaces an earlier draft that held the 2023 ratio fixed and consequently overstated the 2010s by 20–30 percent; that fixed-ratio construction is no longer used.

A stronger check tests the interpolation against numbers it was not given. In a leave-one-anchor-out holdout, we drop each interior anchor, re-interpolate θ from the remaining anchors, predict the held-out year, and compare to the source's published value (Table 2). Across eleven held-out anchors the mean absolute error is 2.4 percent, with a worst case of 5.0 percent (Pew's 2021, the pandemic dip). That is direct evidence that θ moves smoothly enough to interpolate.

Table 2: Leave-one-anchor-out validation of the θ-interpolation step

Series	Held-out year	Published (millions)	Predicted (millions)	Error
U.S., Pew	2015	11.00	10.78	-2.0%
U.S., Pew	2017	10.50	10.79	+2.8%
U.S., Pew	2019	10.20	10.44	+2.3%
U.S., Pew	2021	10.50	11.02	+5.0%
U.S., Pew	2022	11.80	12.00	+1.7%
U.S., Warren/CMS	2019	10.35	10.70	+3.4%
U.S., Warren/CMS	2021	10.30	10.48	+1.8%
U.S., Warren/CMS	2022	10.90	11.06	+1.5%
U.S., Warren/CMS	2023	12.24	12.59	+2.9%
Texas, Warren/CMS	2018	1.73	1.75	+1.1%
Texas, Warren/CMS	2023	2.05	2.10	+2.5%

Mean absolute error 2.4%; median 2.3%.

Source: Each interior anchor is dropped, predicted from the others, and compared to the source's published value. Texas 2036 analysis.

2.3 Texas Insurance Estimate

Let $r^{NC}_{TX,t}$ be the ACS B27020 noncitizen uninsured rate. The Texas undocumented uninsured rate transports that rate on the *odds* scale, calibrated so that 2023 exactly equals MPI's benchmark rate of $1{,}306{,}000/1{,}966{,}000 = 66.4$ percent:

$$\kappa = \frac{\mathrm{odds}(0.664)}{\mathrm{odds}(r^{NC}_{TX,2023})}, \qquad \mathrm{odds}(r) = \frac{r}{1-r}$$

$$\hat{r}^{U}_{TX,t} = \frac{\kappa \times \mathrm{odds}(r^{NC}_{TX,t})}{1 + \kappa \times \mathrm{odds}(r^{NC}_{TX,t})}$$

$$\widehat{Uninsured}^{A}_{TX,t} = \hat{U}^{A}_{TX,t} \times \hat{r}^{U}_{TX,t} \qquad \widehat{Insured}^{A}_{TX,t} = \hat{U}^{A}_{TX,t} - \widehat{Uninsured}^{A}_{TX,t}$$

Why the odds scale. An earlier draft multiplied the noncitizen rate by a constant ratio ($\hat{r}^{U} = \min(1, k\, r^{NC})$). Because pre-ACA noncitizen uninsured rates already exceeded 60 percent, that version produced implausible 85–89 percent undocumented uninsured rates for 2009–2013. Odds-scale calibration is the standard way to transport a rate to a different level: it is bounded inside (0,1), treats insured and uninsured shares symmetrically, and matches the 2023 anchor exactly. It yields 79 percent for 2009 and 64 percent for 2024 (the ratio version is retained in the data as `undoc_uninsured_rate_ratiocal` for comparison). Calibrated rates before roughly 2014 remain more uncertain than recent years because the ACA changed coverage among lawful noncitizens much more than among undocumented immigrants.

No U.S. insurance estimate in the Pew/MPI vintage. There is no published national undocumented insurance benchmark comparable to MPI's Texas figure, and using the raw ACS noncitizen uninsured rate (29 percent in 2024) would badly understate undocumented uninsurance — Warren/CMS's directly published 2024 national figure is 45 percent. The Pew/MPI vintage therefore reports Texas insurance only; the Warren/CMS vintage carries Warren/CMS's direct 2024 U.S. insurance values.

Why this does not require ACS PUMS. For the top-line Texas insured and uninsured counts, the published ACS table is enough because B27020 already reports aggregate noncitizen insurance counts for Texas. PUMS would become necessary if the goal were to model likely legal status or uninsured rates by subgroup, such as age, origin, year of entry, income, education, or geography below the published table level. In other words, the aggregate estimate can use B27020 plus MPI/CMS benchmarks; subgroup estimates would need person-level ACS PUMS.

2.4 Warren/CMS Vintage of the Same Residual Method

The second vintage applies the identical reduced-form residual method to a different published anchor set: the Center for Migration Studies' Warren-based residual estimates. For the United States these are Warren's Journal on Migration and Human Security series and the 2024 data-tool release, giving anchors at 2010 (11,725,000), 2019 (10,350,000), 2021 (10,300,000), 2022 (10,900,000), 2023 (12,244,500), and 2024 (14,611,000). Because these are genuinely different residual-method releases and not derived from the Pew values, the U.S. Warren/CMS line and the U.S. Pew line trace independent histories — Warren/CMS runs higher through the mid-2010s, Pew's 2022–2023 revisions lift the Pew line above Warren/CMS in 2023, and the two nearly meet at 14.6–14.9 million in 2024 (Figure 2). Their disagreement measures methodological variation across two independent implementations of the Warren accounting, not sampling variance in a single estimator.

For Texas, CMS's published "Undocumented Immigrants by State and Year, 2010–2019" dataset supplies anchors at 2016 (1,597,000) and 2018 (1,730,000); the 50-state table adds 2023 (2,052,500) and the data tool adds 2024 (2,317,000). The Texas Warren/CMS line therefore passes through published Warren/CMS values across 2016–2024, and only the pre-2016 years extrapolate the 2016 anchor. Adding CMS's 2010–2015 state values (available in the same dataset) would remove even that short extrapolation.

For insurance in 2024, the CMS CSV reports 1,385,000 uninsured (60 percent) and 932,000 insured undocumented Texans, and nationally 6,575,000 uninsured (45 percent) and 8,036,000 insured. Earlier Warren/CMS-vintage insurance years use the ACS/MPI odds-calibrated rate movement because the current CMS download is a 2024 cross-section.

3 Findings

3.1 The National and Texas Counts Move With ACS Noncitizen Counts

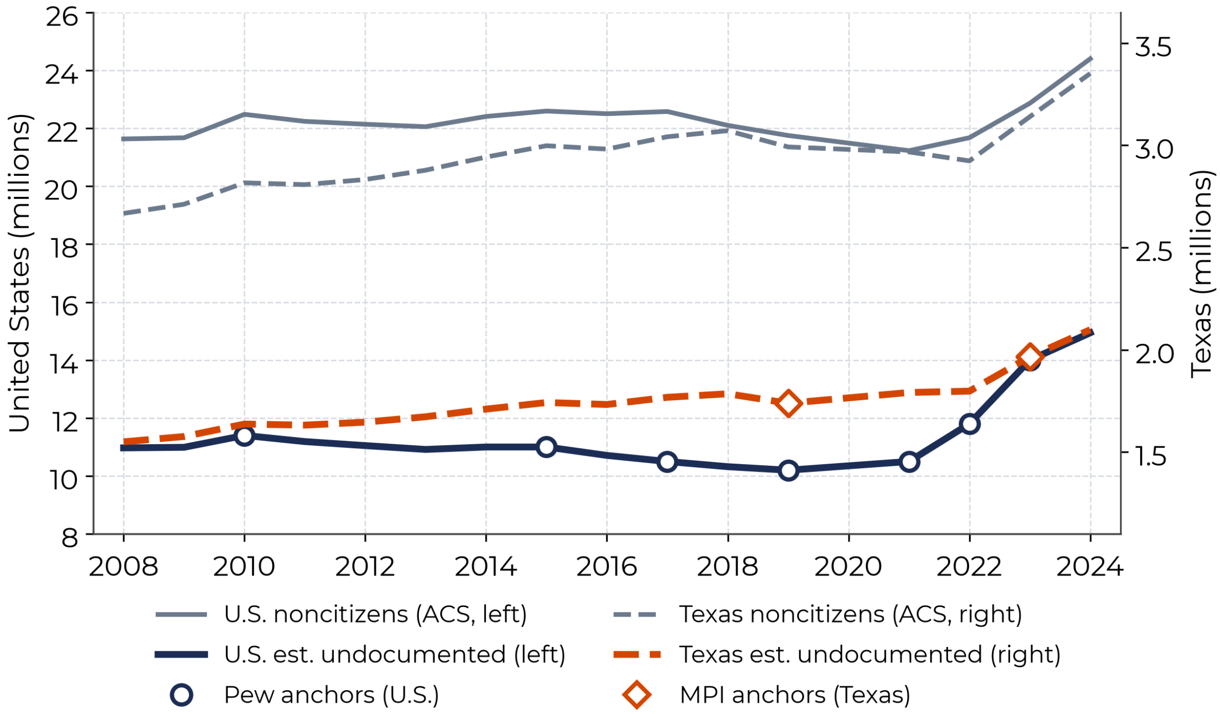

Figure 1: Estimated undocumented population, United States and Texas. Hollow markers are the published Pew (U.S.) and MPI (Texas) anchor values; the estimated lines pass through them by construction (Section 2.1) and interpolate ACS movement between anchors.

Source: Texas 2036 analysis of ACS B05001 via `getcensus`; Pew/MPI vintage of the reduced-form residual method (θ_t pinned to Pew's U.S. and MPI's Texas published residual estimates). Lines connect across 2020, for which no standard 1-year ACS exists. Pew published values from the August 2025 Pew report (2022 revised to 11.8 million).

The Pew/MPI vintage now traces the published national residual-method history: a decline from 11.4 million (2010) to 10.2 million (2019), then a steep climb to 14.0 million by 2023 and an extrapolated 14.9 million in 2024. The hollow markers in Figure 1 are the published Pew (U.S.) and MPI (Texas) values, and the estimated lines pass through each one by construction; only the years between and beyond anchors are filled with observed ACS noncitizen movement. Note that the undocumented and ACS-noncitizen lines no longer share an identical shape — the undocumented share of noncitizens (θ_t) genuinely rose over the period, consistent with Warren et al.'s finding that legal noncitizen inflows slowed relative to total noncitizen inflows — which is the improvement the anchor method delivers over a fixed ratio. In Texas, the Pew/MPI vintage puts 2024 at 2.10 million, or 6.7 percent of the state population, extrapolated forward from the 2023 MPI anchor of 1.97 million.

3.2 Two Independent Residual-Method Vintages Agree on 2024, Differ on History

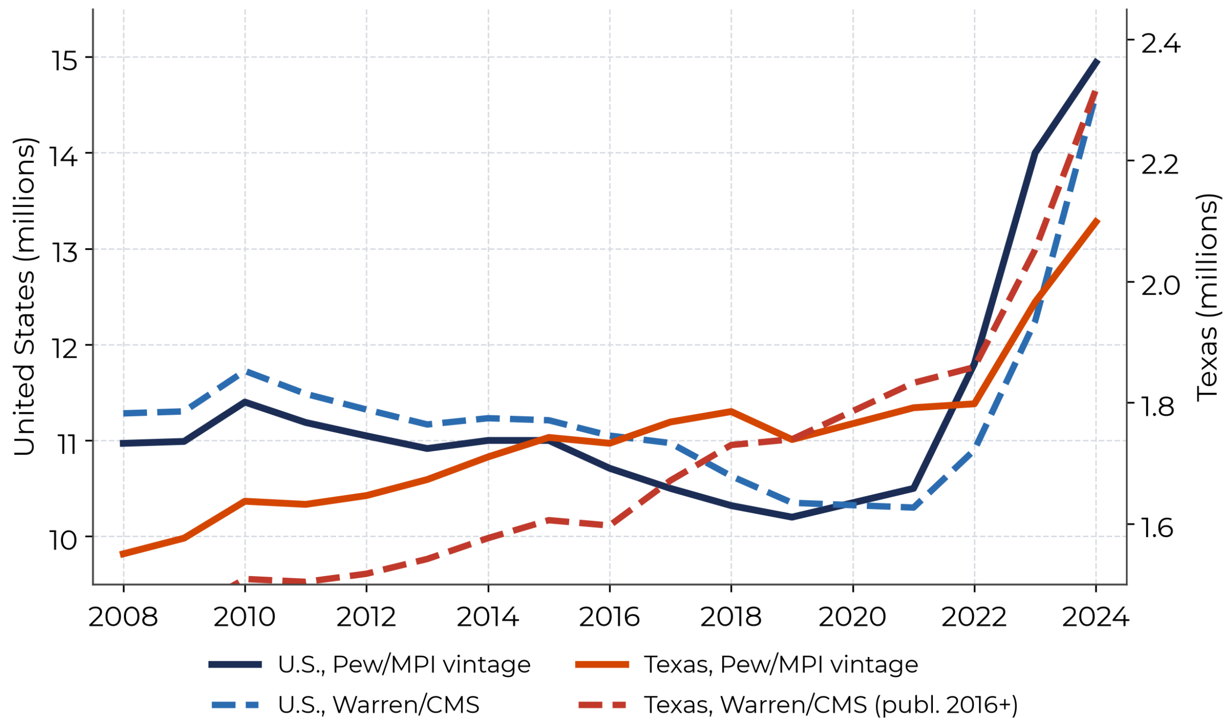

Figure 2: Two vintages of the same reduced-form residual method: Pew U.S. / MPI Texas anchors versus Warren/CMS anchors. Both apply $\widehat{U}_{g,t} = \theta_{g,t} \times C_{g,t}$ to identical ACS $C_{g,t}$; they differ only in whose published U is imported at anchor years.

Source: Texas 2036 analysis of ACS B05001; Pew, MPI, and Warren/CMS published residual-method series.

Because the two vintages follow different published residual-method sources rather than share a 2023 scalar, the distance between them measures genuine disagreement between two independent implementations of the Warren accounting. Nationally, Warren/CMS runs above Pew through the mid-2010s (for example 11.7 versus 11.4 million in 2010), but Pew's 2022–2023 revisions push the Pew line above Warren/CMS (14.0 versus 12.2 million in 2023); the two nearly converge at 14.6–14.9 million in 2024. For Texas the Warren/CMS line is published Warren/CMS history in its own right, tracing 1.60 million (2016) and 1.73 million (2018) before rising to 2.05 million (2023) and 2.32 million (2024). In 2023 the main Texas estimate is 1.97 million, about 4 percent below the Warren/CMS and Pew Texas figures near 2.05 million and equal to the MPI value by construction; in 2024 it is about 9 percent below Warren/CMS's direct 2.32 million, which likely reflects Warren/CMS's fuller year-by-year treatment of liminal statuses, undercount, mortality, emigration, and legal-admission components. The agreement of two independent residual-method constructions on a 2024 U.S. total near 14.6–14.9 million is more informative than either point alone.

3.3 Nationwide, 2024: How the Two Methods Compare Across All States

The same residual logic scales to every state for the latest year. As a national check, we compare two methods for 2024. Method 1 is the reduced form applied with a single national undocumented-share-of-noncitizens ($\theta_{\text{nat}} = 0.599$, calibrated to the CMS national total) multiplied by each state's ACS noncitizen count. Method 2 is CMS's directly published state residual estimate, which lets θ vary by state. Because Method 1 is calibrated to the same national total, the gap between the two is purely the cost of assuming θ is uniform across states.

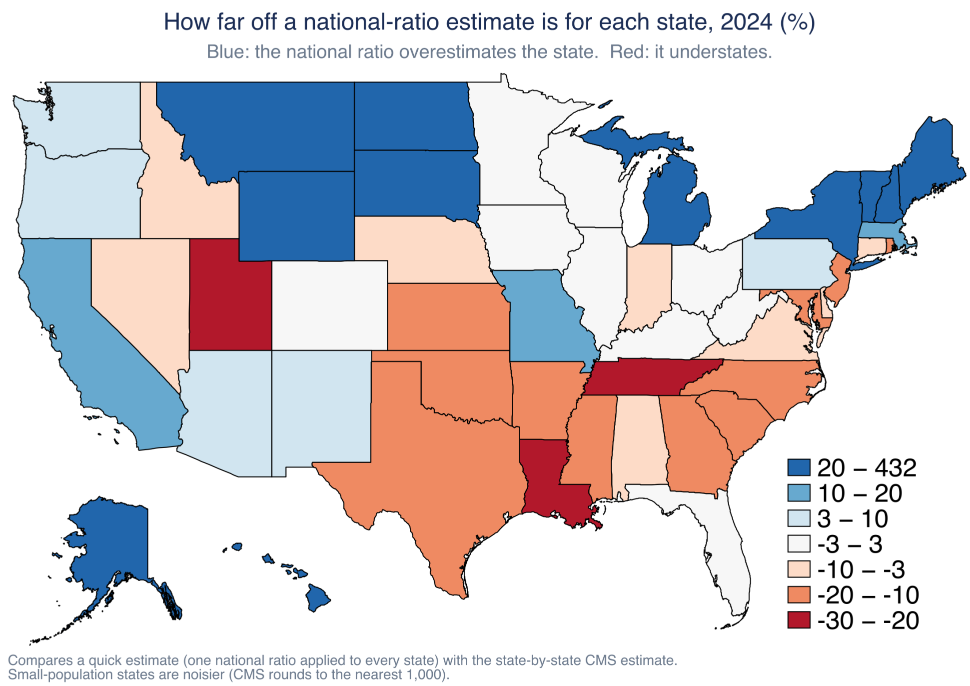

Figure 3: How far off a national-ratio estimate is for each state, 2024, percent. Blue states are overestimated by the national-ratio shortcut; red states are underestimated, relative to the CMS state-by-state estimate.

Source: Texas 2036 analysis of ACS 2024 table B05001 (all states) and the CMS 2024 State and National Data Tool. Small-population states are noisier because CMS rounds to the nearest thousand.

The national-ratio shortcut is accurate for the country as a whole and for much of it state by state: it lands within 10 percent of the CMS state estimate in 22 of 51 states and within 20 percent in 36 of 51. The states where it misses are not random, and the misses are large and one-directional (Figure 3). The shortcut runs too high in gateway and high-legal-immigration states, where many noncitizens are green-card holders, students, or temporary workers (New York +22 percent, California +14 percent), and too low in newer Southern and Mountain destinations, where the undocumented share of noncitizens runs high (Louisiana −30 percent, Tennessee −23 percent, Utah −23 percent, Texas −13 percent). The reason is simple: the shortcut assumes every state has the same undocumented share of noncitizens, and that share is genuinely higher in the newer destinations and lower in the gateways.

This matters because the direction and size of the error are predictable. An analyst who used the national ratio for Louisiana or Tennessee would understate its undocumented population, and with it the uninsured count and the scope of any coverage program, by roughly a quarter to a third. Texas is the clearest case for local work: the national ratio understates it by about an eighth, which is why the Texas estimates in this memo use the state-specific MPI and CMS anchors rather than the shortcut. The value of the map is practical: it tells an analyst which states a quick national estimate can be trusted for and which need a state-specific number first. A fuller treatment of the all-states method, with applied examples, is in the companion manuscript.

How uncertain are the state numbers? Two different kinds of uncertainty, which we keep separate. ACS *sampling* error is a genuine margin, and for these estimates it is small (about 1 to 2 percent for large states). The larger and more important uncertainty is disagreement among the published sources — how far apart CMS, Pew, and MPI are for the same state — which we report as a *range*, not a confidence interval. Three estimates from different methods and vintages are not a sample from a distribution, so their spread is a robustness range, not a variance to convert into a bound. That range is wide and uneven (Figure 4): the three sources span a median of roughly a quarter of their mean across states, from near-agreement in Texas (1.97 to 2.05 million) to a 2.5-fold gap in Michigan (0.08 to 0.20 million); nationally they span 12.2 to 14.0 million. Our shortcut lands inside the published range in about half the states, missing in the direction the map flags. For a state number, which credible source you anchor on matters more than survey sampling.

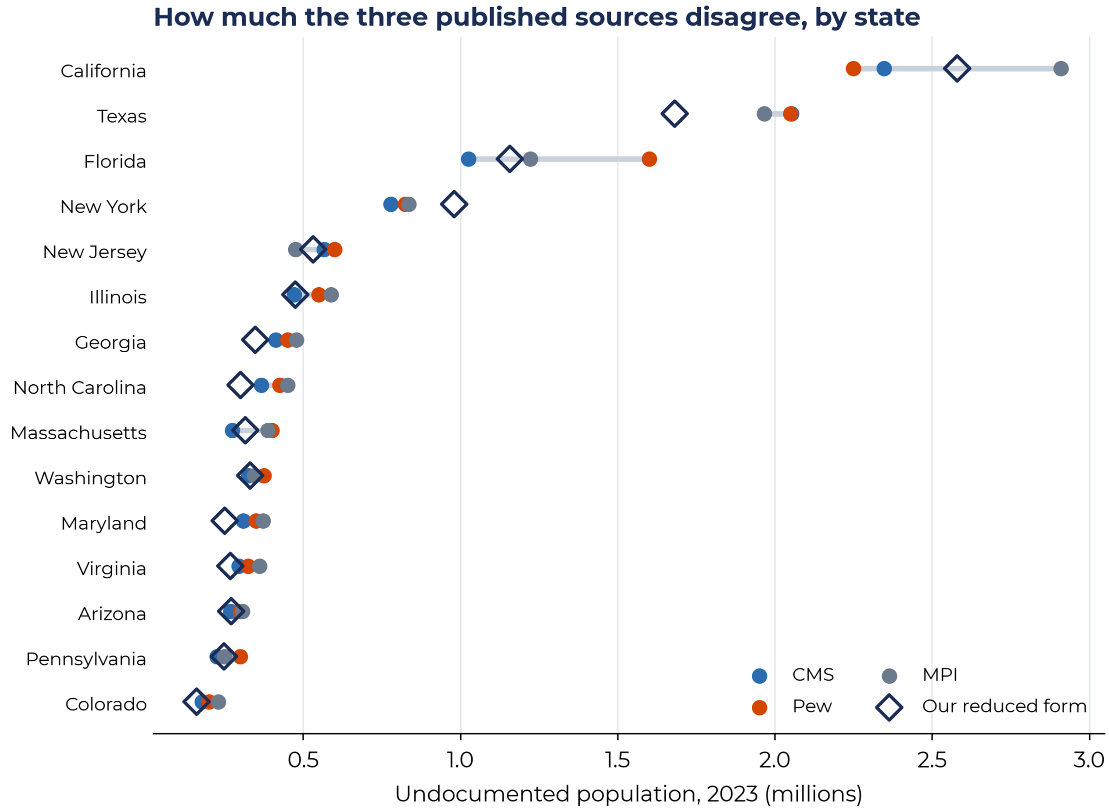

Figure 4: The three published residual-method sources (CMS, Pew, MPI) for 2023, by state, with our reduced-form estimate for comparison. The gray bar is each state's lowest-to-highest source range.

Source: Immigration Research Initiative 50-state table (2023); Texas 2036 analysis.

The spread is widest where the undocumented are a *small* share of a state's noncitizens. Because the method estimates them as noncitizens minus the legal population, Texas (where they are about two-thirds of noncitizens) is robust to how each source pegs the legal count, while a state like Michigan (under half, in a noncitizen pool heavy with students, work-visa holders, and refugees) sees the same small differences swing a much thinner residual. Sampling error is small everywhere; the width is cross-source disagreement, and it is largest for low-share, dispersed states. Texas sits near the floor of that agreement, not the norm — the median state's three sources span about a quarter of their mean.

One honest caveat on the width: the cross-source term is accurate but somewhat inflated as a gauge of *pure method* disagreement, because the three sources are not estimating the same-year quantity. CMS and Pew both use the 2022 ACS, while MPI pools 2019–2023 ACS weighted to 2023, so part of each state's gap reflects different reference periods and smoothing rather than a real disagreement about the 2023 population. Sources on a shared vintage would agree more closely (Texas, where all three align, is a hint). We report the full spread because it answers the question analysts face — how much the number depends on which source you pick — but the widest state ranges reflect vintage *and* method, not method alone.

3.4 Texas Insurance Estimates

Table 3: Texas estimated undocumented population and insurance status

Year	ACS noncitizens (millions)	Pew/MPI estimate (millions)	Undoc. share of TX pop.	Warren/CMS check (millions)	Insured (millions)	Uninsured (millions)	Uninsured rate among undoc.
2010	2.82	1.64	6.5%	1.51	0.35	1.29	79%
2015	3.00	1.74	6.3%	1.61	0.54	1.20	69%
2019	2.99	1.74	6.0%	1.74	0.54	1.20	69%
2021	2.97	1.79	6.1%	1.83	0.57	1.23	68%
2022	2.92	1.80	6.0%	1.86	0.59	1.21	67%
2023	3.14	1.97	6.4%	2.05	0.66	1.31	66%
2024	3.35	2.10	6.7%	2.32	0.76	1.34	64%

Source: Texas 2036 analysis of ACS B05001 and B27020 via `getcensus`. Pew/MPI vintage of the reduced-form residual method uses MPI Texas anchors (2019, 2023). Insurance values begin in 2009; selected years shown. The Pew/MPI Texas line equals MPI's published residual estimate in 2019 and 2023 and interpolates ACS movement elsewhere; 2010–2018 extrapolate the 2019 anchor. The Warren/CMS Texas column uses published Warren/CMS state anchors at 2016, 2018, 2023, and 2024. Insured/uninsured columns use the odds-calibrated rate described in Methods.

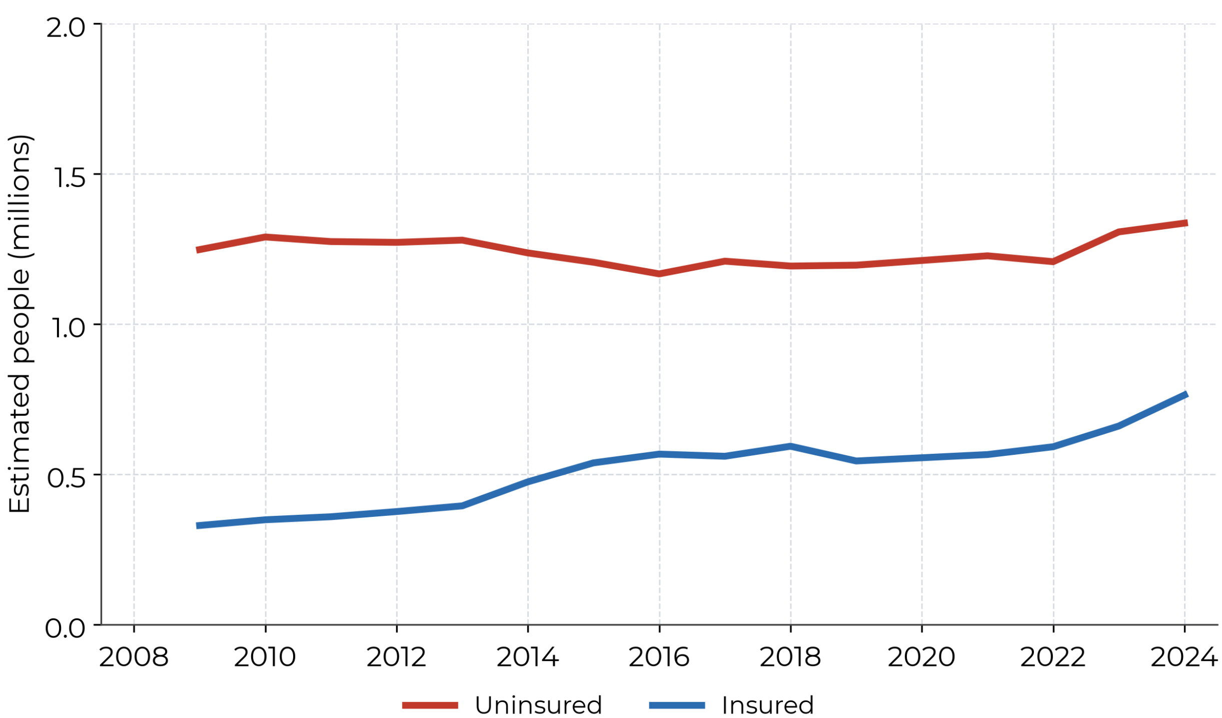

Figure 5: Estimated insured and uninsured undocumented Texans
Source: ACS B27020 via `getcensus`; MPI 2023 Texas unauthorized profile.

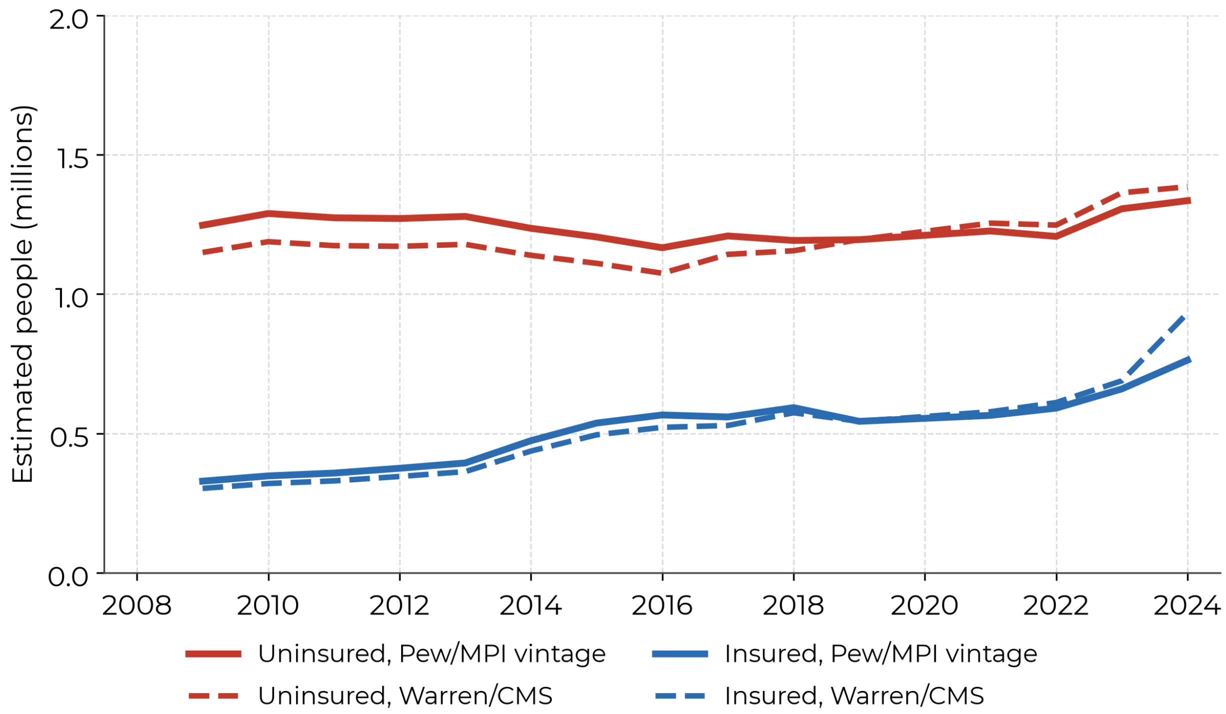

Figure 6: Texas insurance counts under two residual-method vintages: Pew/MPI vintage and Warren/CMS

Source: Texas 2036 analysis. CMS population anchors are published state data (2016, 2018, 2023, 2024); 2024 insurance split is the direct CMS count.

The Pew/MPI Texas vintage puts 2024 undocumented uninsured Texans at roughly 1.34 million, or 64 percent of estimated undocumented Texans and 4.3 percent of the total Texas population. The uninsured count stays near its 2023 level even as the estimated undocumented population grows because the ACS noncitizen uninsured rate fell in 2024, pulling the calibrated undocumented rate down from 66 to 64 percent. The estimated insured undocumented count rises to about 0.76 million, or 36 percent of estimated undocumented Texans. The direct Warren/CMS 2024 value lands close by: about 1.39 million uninsured (60 percent) and 0.93 million insured. Two independent residual-method vintages putting Texas undocumented uninsurance at 1.3–1.4 million is the most robust finding in this memo.

The insurance transport is calibrated only once, to MPI's 2023 Texas benchmark, so it is fair to ask whether that single calibration travels. Applying the one Texas-calibrated constant to every state's ACS noncitizen uninsured rate and comparing to CMS's directly published state uninsured rate (Figure 7) shows it does reasonably well: the points track the 45-degree line (correlation 0.71; population-weighted mean error about 3.7 percentage points), with the expected scatter for small states. This checks the rate transport, not a second data source.

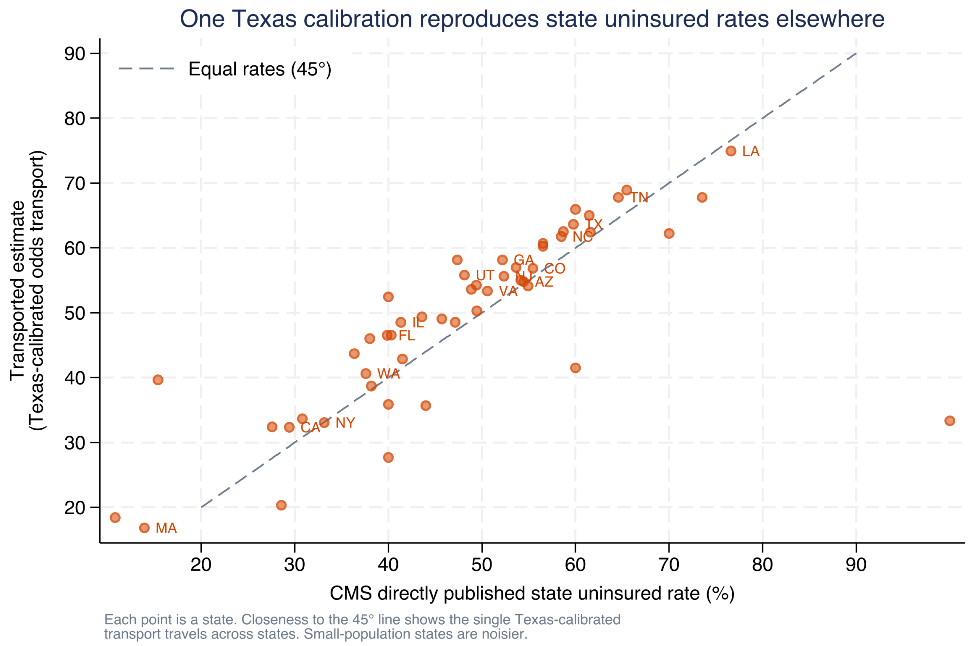

Figure 7: The single Texas-calibrated odds transport applied to every state, against CMS's directly published state uninsured rate, 2024. Closeness to the 45-degree line shows the calibration travels across states.

Source: Texas 2036 analysis of ACS 2024 B27020 and the CMS 2024 State and National Data Tool.

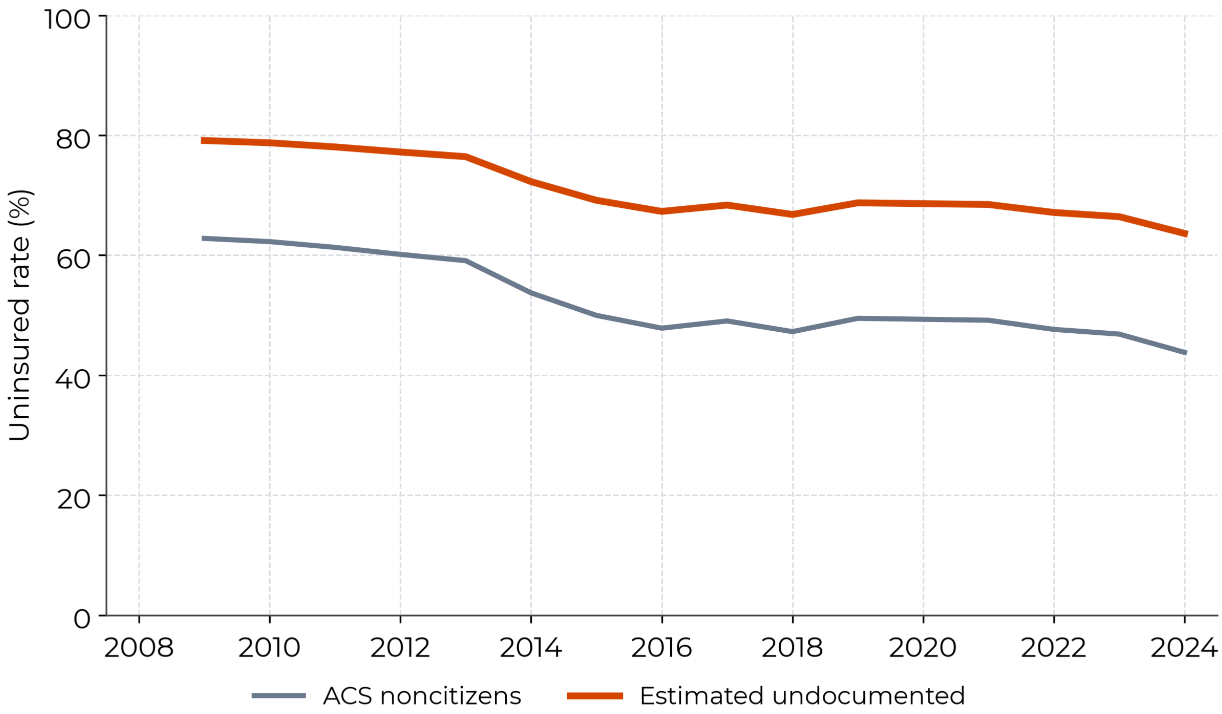

Figure 8: ACS noncitizen uninsured rate and benchmarked undocumented uninsured rate
Source: Texas 2036 analysis of ACS B27020; undocumented line odds-calibrated so 2023 equals MPI's 66 percent Texas uninsured benchmark. Pre-2014 calibrated rates are more uncertain (see Methods).

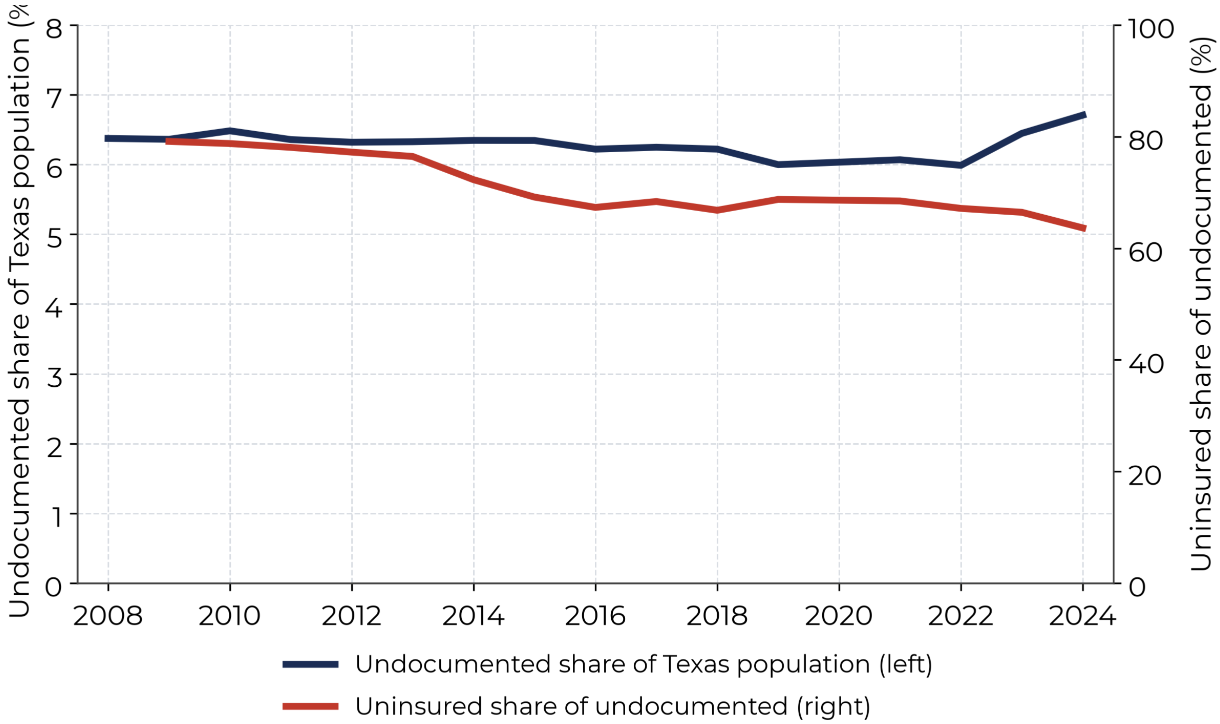

Figure 9: Texas percentage measures: undocumented share of the state population and uninsured share of the undocumented. (The insured share is omitted: it is exactly 100 minus the uninsured share.)

Source: Texas 2036 analysis of ACS B05001 and B27020; population series anchored to MPI Texas 2019 and 2023; insurance odds-calibrated to MPI 2023.

3.5 Going to Subgroups: What the Microdata Add, and the Cost

The published tables give totals. When a decision needs a subgroup or a smaller area, the same method extends to the ACS Public Use Microdata Sample (PUMS): tabulate noncitizen counts by cell from the person file, then apply the same undocumented share and insurance transport used above. These are transported estimates, not person-level legal-status determinations. Running this on the 2024 Texas PUMS reproduces the statewide numbers and then breaks them out, for example about 429,000 undocumented adults living with a child under six (roughly 68 percent uninsured), or about 46,000 in the Rio Grande Valley living with children (76 percent uninsured, well above the state's 64 percent).

The trade is precision. Figure 10 shows the ACS sampling margin, which widens as the group shrinks, from about ±2 percent statewide to about ±20 percent for undocumented children in a single county. Subgroup detail is available and useful for targeting, but the smaller the cell, the more it should be read as a range rather than a point (and cross-source and undercount uncertainty sit on top of the sampling margin shown here).

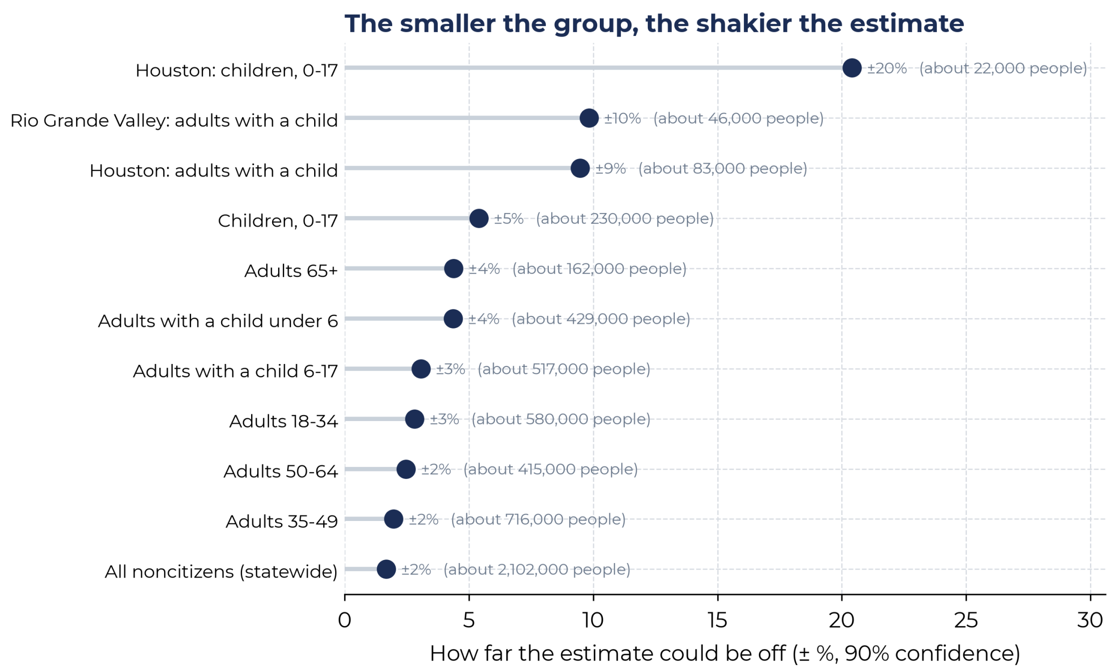

Figure 10: The smaller the group, the shakier the estimate. Each bar is the 90 percent margin of error on a Texas undocumented estimate from the 2024 ACS PUMS, combining sampling error and the cross-source spread in the undocumented share of noncitizens.

Source: Texas 2036 analysis of the 2024 ACS 1-year PUMS for Texas; margins from the 80 replicate weights plus the 2023 cross-source spread. Transported estimates, not person-level legal-status imputations.

4 Caveats and Next Steps

Interpretation cautions

- **Legal status is modeled, not observed.** ACS noncitizens include lawful permanent residents, lawful temporary migrants, refugees, asylees, people with liminal status, and undocumented residents. "Undocumented" here includes liminal statuses, matching all three residual-method sources.
- **The workflow implements the residual accounting indirectly.** We use ACS C_t directly and inherit each anchor source's implicit L_t via its published U_t; we do not rebuild lawful admissions, mortality, emigration, or undercount from primary sources. Warren et al.'s (2023) year-by-year approach is the natural full-strength alternative and the accompanying Stata package supports it directly.
- **Each series equals its residual-method source at anchor years; the tails are extrapolations.** U.S. 2008–2009 and Texas 2008–2018 extend the nearest anchor for the Pew/MPI vintage and are not published values; treat those early Texas years as indicative (Section 2.1).
- **A methodology break sits at 2022–2023.** Part of the sharp rise reflects Pew's reweighting and CMS/Warren's new 2024-release methodology, not only real population growth, so change measured across that break is not purely demographic.
- **The Texas Warren/CMS vintage is published Warren/CMS state data for 2016–2024** (2016, 2018, 2023, 2024 anchors); only pre-2016 extrapolates. The U.S. Warren/CMS vintage likewise uses Warren/CMS's published annual estimates.
- **Insurance estimates lean on one anchor.** Texas rates move with ACS noncitizen rates and are odds-calibrated to a single MPI 2023 benchmark; pre-2014 calibrated rates are more uncertain because the ACA changed lawful-noncitizen coverage much more than undocumented coverage. No U.S. main-series insurance estimate is produced.
- **ACS survey error and undercount matter.** The workflow retains margins of error in raw files, but the published output does not propagate uncertainty through every modeled assumption.

Recommended next improvements:

- **Implemented in this revision:** each residual-method vintage now interpolates its own source's published anchors — Pew U.S. (2010–2023), MPI Texas (2019, 2023), Warren/CMS U.S. (2010–2024), and Warren/CMS Texas (2016, 2018, 2023, 2024) — rather than holding a single 2023 ratio fixed. Each vintage reproduces its residual-method source's published history at its anchor years.
- **Implement Warren et al. (2023) directly.** Extend the workflow with a year-by-year

construction of L_t from DHS lawful admissions by class and cohort, life-table survival, ACS-derived emigration, and age-, sex-, Hispanic-origin-specific undercount adjustments. That would expose (and permit sensitivity tests on) the demographic components that currently enter only through the anchor sources. The companion `residualundoc.ado` package supports this via its `direct` subcommand.

- Add the CMS 2010–2015 Texas state values (same published Warren/CMS dataset) to remove the short pre-2016 Warren/CMS Texas extrapolation, and locate archived MPI Texas profile vintages to give the Pew/MPI Texas line a pre-2019 anchor (it currently extrapolates 2008–2018 from the 2019 MPI value).
- Re-verify the digit-level MPI Texas 2019 value (1,739,000) against an archived MPI profile before external release; the current figure is corroborated only at the "about 1.7 million" level by secondary reporting.
- Corroborate the Texas insurance anchor against KFF's coverage-by-status estimates and CMS's 2024 Texas insurance split; consider presenting a benchmark band (MPI-calibrated and CMS-direct) instead of a single line.
- **Done this revision (extendable):** a worked ACS PUMS disaggregation to Texas subgroups and sub-state regions (Figure 10). A natural extension models likely legal status per record — by age, origin, year of entry, education, and income — with multiple-imputation variance, rather than transporting a single share.
- **Partly implemented in this revision:** state estimates now carry a 90 percent ACS *sampling* interval, with cross-source disagreement reported separately as a range rather than folded into it (Findings). Still open: propagating the undercount adjustment and insurance-rate sensitivity into those bounds, and a single interval spanning every modeled step.

5 Reproducibility

The full estimation workflow is in `code/01_acs_residual_undocumented_estimates.do`. It writes raw ACS and CMS pulls to `data/raw`, processed panels to `data/processed`, and figures to `figures`. A Python mirror, `code/02_rebuild_outputs_python.py`, reproduces the processed outputs, table, and figures from the saved raw files for environments without Stata. The all-states estimate, intervals, and calibration are in `code/04_state_comparison_2024.do` (which calls the companion `residualundoc apply`); the holdout validation is in `code/07_holdout_validation.py`; the PUMS subgroup example is in `code/06_pums_texas_2024.py`. Method assumptions and benchmark provenance are logged in `data/processed/method_assumptions.csv`; the published anchors and the (zero) deviation of the estimate from them at anchor years are in `data/processed/anchor_validation.csv`. A generalized Stata implementation of the reduced-form, cross-sectional, direct, transport, and comparison steps lives in `stata_package/residualundoc.ado` (with `residualundoc.sthlp` and a self-contained example).

Sources (all accessed July 2026). ACS 1-year tables B05001 and B27020 and the 2024 PUMS, U.S. Census Bureau (api.census.gov and census.gov PUMS); CMS State and National Data Tool (data.cmsny.org, methodology at /about); Pew Research Center (Aug. 2025 report and methodology); Migration Policy Institute (methodology and Texas profile); Immigration Research Initiative, *50 States: Immigrants by Number and Share*, Nov. 2025 (immresearch.org).